



College of Science and Technology
School of Science
Department of Mathematics

Module name: Mathematics for Engineers I (MATH I)

Module code: MAT1166

Academic year: 2024-2025

Groups: Year I ETE, EPE and Computer Engineering

Lecturer: Justine DUSHIMIRIMANA

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MODULE DESCRIPTION

1. Module Title: MATHEMATICS FOR ENGINEERS I (MAT 1166).
2. Level: 1 Semester: 1 Credits: 10.
3. Pre-requisite or co-requisite modules, excluded combinations. Pre-requisite:
Pass A level Mathematics
4. **Brief description of aims and content**

The aim is to provide a link between mathematics that required for Engineering. A review unit is followed by units on calculus on functions of one real variable, numerical and functions series, introduction to matrices theory, and element of vector's spaces.

5. LEARNING OUTCOMES

A. Knowledge and Understanding

Having successfully completed the module, students should be able to demonstrate knowledge and understanding of:

- A.1. The basics of calculus on functions of one variable, functions series, vectors and matrices.
- A.2. The implications of the basic mathematical theories.

B. Cognitive/Intellectual skills/Applications of Knowledge

Having successfully completed the module, students should be able to:

- B.1. Present simple arguments and conclusion using the mathematical theories.
- B.2. Analyze and evaluate problems in mathematical and engineering.

C. Communication /ICT/Numeracy/Analytic Techniques/Practical Skills.

Having successfully completed the module, students should be able to:

- C.1. Apply the mathematical knowledge to solve problems in range of engineering situations.
- C.2. Carry out mathematical and numerical manipulation with confidence and accuracy.

D. General transferable skills

Having successfully completed the module, students should be able to:

- D.1. Assimilate Abstract Ideas;
- D.2. Communicate information having a mathematical content.

6. INDICATIVE CONTENTS

Introduction: Recall of some background tools: sets of real numbers, interval, numerical function and their classical property.

CHAPTER I: Calculus and functions of one real variable

- I.1. Definitions of limits and continuity, techniques of finding limits.
- I.2. Techniques of differentiation, tangent lines and rates of change.
- I.3. Mechanical and geometric Applications of derivatives. Indeterminate form and L' hospital's rule
- I.4. Techniques of indefinite integration.
- I.5. Definite integrals, properties of definite integrals.
- I.6. Mechanical and geometric Applications of definite integrals.
- I.7 Improper integrals.

Chapter II: Numerical and functional series.

- II.1. Convergence and divergence of numerical sequences and series,
- II.2. Techniques for testing the convergence of numerical sequences and series.
- II.3. Power series, Maclaurin series, Taylor series and their applications.

Chapter III. Introduction to matrices theory.

- III.1. Introduction to matrices, Algebra of matrices, Special matrices. Determinants and their properties.
- III.2. System of linear equations, solution by Elimination Methods (Gauss-Elimination, Gauss-Jordan algorithms)

Chapter IV. Element of Vector's spaces

- IV.1. Vectors, operation by vectors, scalar, vector and triple products.
- IV.2. Linear independence, bases, Vector space.
- IV.3. Linear operators, Eigenvalues, Eigenvectors; Orthogonal matrices.

$$7. \text{ ASSESSMENT STRATEGY: } \left\{ \begin{array}{l} \text{Quizzes: 10} \\ \text{Assignments: 10} \\ \text{CAT (2): 30} \\ \text{Final Exam: 50} \end{array} \right.$$

8. . REFERENCES:

1. Robert A. Adams and Christopher Essex, Calculus: A Complete Course,

- 7th Edition, Pearson Education Canada, 2010.
2. David C. Lay, Linear Algebra and Its Applications, 4th Edition,
 3. Linear Algebra with applications; sixth Edition, Steven J. Leon; 2002 by Printice-Hall, Inc.
 4. Mathematics for science, S.M. UPPAL, 1996, New Age International(P), Limited, publishers.